

Qatar-1b

R-1677

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_161114 · created 2026-08-02T11:12:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457706.66842 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.054 +/- 0.046
Transit depth (Rp/Rs)^2	0.29 +/- 0.5 [%]
Orbital Inclination (inc)	80.22 +/- 3.07
Airmass coefficient 1 (a1)	0.916 +/- 0.025
Airmass coefficient 2 (a2)	0.0555 +/- 0.0075
Scatter in the residuals of the lightcurve fit is	2.71 %
Best Comparison Star	#3 - [518, 192]
Optimal Method	PSF photometry
Transit Duration (day)	0.023 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.054 $\pm$ 0.046 vs 0.1463 (deviation 1.41 σ)
Duration fit vs published	0.023 d vs 0.0692 d (deviation 1.24 σ)
Mid-time O-C	-0.1 min
Depth SNR	0.8
Residual scatter	2.71 %

Light curve

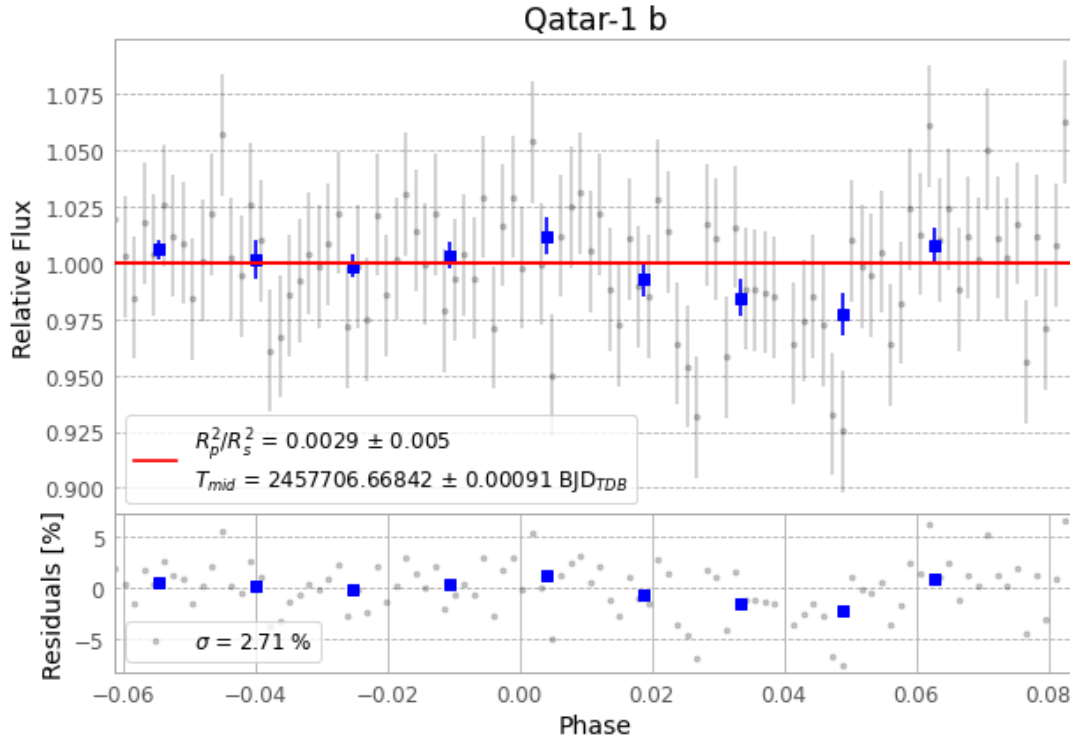


Figure 1 — FinalLightCurve_Qatar-1 b_2016-11-13.png (SHA-256 922a54d2c304aba2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [212, 265], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fb6b2358896bd18c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

100 FITS frames (66 MB), Qatar-1161114015412.FITS → Qatar-1161114065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-161114.log (0d26bd813001...), FinalLightCurve_Qatar-1 b_2016-11-13.png (922a54d2c304...), FinalLightCurve_Qatar-1 b_2016-11-13.csv (0d82b37fc65f...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 11:12Z.